



PFAI

Lab Assignment # 3



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SP24-BAI-027

Task 1 Version Control:

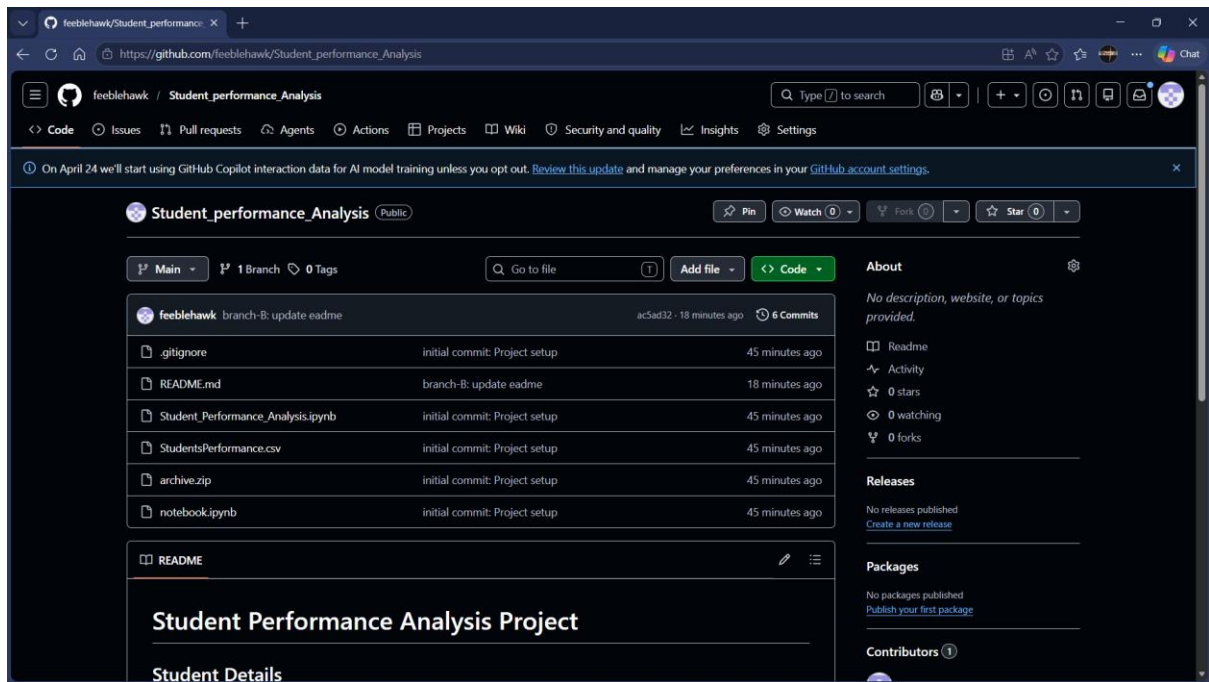
Commands and Sample Outputs:

Repo url: [feebleshawk/Student_performance_Analysis](https://github.com/feebleshawk/Student_performance_Analysis)

```
Analysis\venv\Scripts\Activate.ps1)
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git init
Initialized empty Git repository in C:/files/PFAI/Student_performance_Analysis/.git/
(.venv) PS C:\files\PFAI\Student_performance_Analysis> echo "data/"> .gitignore
(.venv) PS C:\files\PFAI\Student_performance_Analysis> echo ".ipynb_checkpoints">> .gitignore
(.venv) PS C:\files\PFAI\Student_performance_Analysis> echo "#Student Performance Analysis" > README.md
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config advice.addEmptyPathsSpec false"
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git add .
warning: in the working copy of 'Student_Performance_Analysis.ipynb', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'StudentsPerformance.csv', LF will be replaced by CRLF the next time Git touches it
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git commit -m "initial commit: Project setup"
[master (root-commit) 75f28df] initial commit: Project setup
6 files changed, 1869 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md
create mode 100644 Student_Performance_Analysis.ipynb
create mode 100644 StudentsPerformance.csv
create mode 100644 archive.zip
create mode 100644 notebook.ipynb
(.venv) PS C:\files\PFAI\Student_performance_Analysis>
```

```
1 file changed, 0 insertions(+), 0 deletions(-)
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git checkout master
Switched to branch 'master'
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git merge branch-A
Updating efc8b67..6b2f1a3
Fast-forward
 README.md | Bin 764 -> 826 bytes
1 file changed, 0 insertions(+), 0 deletions(-)
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git merge branch-B
Updating 6b2f1a3..ac5ad32
Fast-forward
 README.md | Bin 826 -> 876 bytes
1 file changed, 0 insertions(+), 0 deletions(-)
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git merge branch-B
Already up to date.
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git add README.md
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git commit -m "eslove merge conflict between branch-A and Branch-B"
On branch master
nothing to commit, working tree clean
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git log --online --graph --all
* ac5ad32 (HEAD -> master, branch-B) branch-B: update eadme
* 6b2f1a3 (branch-A) branch-A: update README
* efc8b67 Merge features to masin(master)
* 31fefd3 (feature/statistics) feature/statistics: Add statistics analysis section
* 2b7b805 updated README.md
* 75f28df initial commit: Project setup
(.venv) PS C:\files\PFAI\Student_performance_Analysis>
```

```
hint: Disable this message with "git config advice.refSyntax false"
(.venv) PS C:\files\PFAI\Student_performance_Analysis> git push -u origin Main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 8 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (21/21), 24.76 KiB | 603.00 KiB/s, done.
Total 21 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), done.
To https://github.com/feebleshawk/Student_performance_Analysis.git
 * [new branch]      Main -> Main
branch 'Main' set up to track 'origin/Main'.
(.venv) PS C:\files\PFAI\Student_performance_Analysis>
```



Common Setup and Import libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set global plot style
sns.set_style('whitegrid')
plt.rcParams['figure.dpi'] = 120

# Load raw dataset
df = pd.read_csv('StudentsPerformance.csv')

print('Dataset loaded successfully!')
print('Shape:', df.shape)
print('\nFirst 5 rows:')
df.head()
```

Output:

```
Dataset loaded successfully!
Shape: (1000, 8)
```

```
First 5 rows:
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

TASK 2 — Descriptive Statistics

Source Code:

```
scores = ['math score', 'reading score', 'writing score']

print('='*60)
print('          TASK 2 – DESCRIPTIVE STATISTICS')
print('='*60)

# ---- Mean, Median, Mode, Std, Min, Max ----
print('\n--- MEAN, MEDIAN, MODE, STD, MIN, MAX ---\n')
for col in scores:
    print(f'{col.upper()}\n')
    print(f'  Mean   : {df[col].mean():.2f}')
    print(f'  Median : {df[col].median():.2f}')
    print(f'  Mode    : {df[col].mode()[0]}')
    print(f'  Std Dev: {df[col].std():.2f}')
    print(f'  Min     : {df[col].min()}')
    print(f'  Max     : {df[col].max()}')
    print()

# ---- Mean vs Median Comparison ----
print('='*60)
print('          MEAN vs MEDIAN COMPARISON')
print('='*60)
print(f'{"Score":<22} {"Mean":>8} {"Median":>8} {"Diff":>8}')
print('-'*50)
for col in scores:
    mean    = df[col].mean()
    median  = df[col].median()
    diff    = mean - median
    print(f'{col:<22} {mean:>8.2f} {median:>8.2f} {diff:>8.2f}')

# ---- GroupBy Gender ----
print('\n' + '='*60)
print('          AVERAGE SCORES BY GENDER')
print('='*60)
print(df.groupby('gender')[scores].mean().round(2))

# ---- GroupBy Parental Education ----
print('\n' + '='*60)
print('          AVERAGE SCORES BY PARENTAL LEVEL OF EDUCATION')
print('='*60)
print(df.groupby('parental level of education')[scores].mean().round(2))
```

```

# ---- GroupBy Test Prep ----
print('\n' + '='*60)
print('          AVERAGE SCORES BY TEST PREPARATION COURSE')
print('='*60)
print(df.groupby('test preparation course')[scores].mean().round(2))

# ---- GroupBy Lunch ----
print('\n' + '='*60)
print('          AVERAGE SCORES BY LUNCH TYPE')
print('='*60)
print(df.groupby('lunch')[scores].mean().round(2))

```

Sample Output:

```

=====
TASK 2 – DESCRIPTIVE STATISTICS
=====

--- MEAN, MEDIAN, MODE, STD, MIN, MAX ---

MATH SCORE
Mean    : 66.09
Median  : 66.00
Mode    : 65
Std Dev: 15.16
Min     : 0
Max     : 100

READING SCORE
Mean    : 69.17
Median  : 70.00
Mode    : 72
Std Dev: 14.60
Min     : 17
Max     : 100

WRITING SCORE
Mean    : 68.05
Median  : 69.00
Mode    : 74
Std Dev: 15.20
Min     : 10
Max     : 100

```

MEAN vs MEDIAN COMPARISON			
Score	Mean	Median	Diff
math score	66.09	66.00	0.09
reading score	69.17	70.00	-0.83
writing score	68.05	69.00	-0.95

AVERAGE SCORES BY GENDER			
	math score	reading score	writing score
female	63.63	72.61	72.47
male	68.73	65.47	63.31

AVERAGE SCORES BY PARENTAL LEVEL OF EDUCATION			
	math score	reading score	writing score
associate's degree	67.88	70.93	69.90
bachelor's degree	69.39	73.00	73.38
high school	62.14	64.70	62.45
master's degree	69.75	75.37	75.68
some college	67.13	69.46	68.84
some high school	63.50	66.94	64.89

AVERAGE SCORES BY TEST PREPARATION COURSE			
	math score	reading score	writing score
completed	69.70	73.89	74.42
none	64.08	66.53	64.50

AVERAGE SCORES BY LUNCH TYPE			
	math score	reading score	writing score
free/reduced	58.92	64.65	63.02
standard	70.03	71.65	70.82

OBSERVATIONS:

- Female students score higher in reading and writing; males score higher in math.
- Higher parental education = higher student scores across all subjects.
- Mean and Median are very close, showing scores are roughly normally distributed.
- Students who completed test prep course score 5-8 points higher on average.
- Students with standard lunch consistently outperform free/reduced lunch students.

TASK 3 — Data Cleaning

Source Code:

```
# Create a messy version of the dataset
df_raw = pd.read_csv('data/StudentsPerformance.csv')
messy = df_raw.copy()

np.random.seed(42)
```

```

# Problem 1: Add missing values
idx_nan = np.random.choice(messy.index, 40, replace=False)
messy.loc[idx_nan[:15], 'math score'] = np.nan
messy.loc[idx_nan[15:28], 'reading score'] = np.nan
messy.loc[idx_nan[28:], 'writing score'] = np.nan

# Problem 2: Add duplicate rows
duplicate_rows = messy.sample(25, random_state=1)
messy = pd.concat([messy, duplicate_rows], ignore_index=True)

# Problem 3: Add outliers (impossible scores)
messy.loc[5, 'math score'] = 150 # above 100
messy.loc[10, 'reading score'] = -5 # below 0
messy.loc[15, 'writing score'] = 999 # absurd value

# Problem 4: Add invalid category values
messy.loc[3, 'gender'] = 'unknown'
messy.loc[7, 'gender'] = 'MALE' # wrong case
messy.loc[20, 'lunch'] = 'N/A'

# Problem 5: Add whitespace in string columns
messy.loc[1, 'gender'] = ' female '
messy.loc[2, 'gender'] = ' male'

# Save messy file
messy.to_csv('data/messy_students.csv', index=False)

print('Messy dataset created and saved!')
print('Messy shape:', messy.shape)

```

Sample Output:

Messy dataset created and saved!

Loaded messy_students.csv

Shape: (1005, 8)

Problems:

-- info() --

<class 'pandas.DataFrame'>

RangeIndex: 1005 entries, 0 to 1004

Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	gender	1004 non-null	str
1	race/ethnicity	1005 non-null	str
2	parental level of education	1005 non-null	str
3	lunch	1005 non-null	str
4	test preparation course	1005 non-null	str
5	math score	1004 non-null	float64
6	reading score	1004 non-null	float64
7	writing score	1005 non-null	int64

dtypes: float64(2), int64(1), str(5)

memory usage: 62.9 KB

Missing values filled

gender	1
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	1
reading score	1
writing score	0

dtype: int64

Duplicates removed

Duplicates remaining: 0

Outliers removed (range: 27.0 - 107.0)

Cleaned file saved to cleaned_students.csv

== BEFORE vs AFTER ==

Metric	Before	After
-----	-----	-----
Rows	1005	990
Columns	8	8
Missing Values	3	2
Duplicates	5	0
Outliers	9	0

Cleaned Data Statistics:

	math score	reading score	writing score
count	990.00	989.00	990.00
mean	66.48	69.44	68.37
std	14.57	14.28	14.82
min	27.00	23.00	15.00
25%	57.00	60.00	58.00
50%	66.00	70.00	69.00
75%	77.00	80.00	79.00
max	100.00	100.00	100.00

TASK 4 — EDA & Correlation

Source Code:

```
# — Load Cleaned Data —————
df = pd.read_csv("outputs/cleaned_students.csv")
print("Shape:", df.shape)
print(df.head())

# — GroupBy: Gender —————
print("\n===== BY GENDER =====")
print(df.groupby('gender').mean(numeric_only=True).round(2))

# — GroupBy: Race/Ethnicity —————
print("\n===== BY RACE/ETHNICITY =====")
print(df.groupby('race/ethnicity').mean(numeric_only=True).round(2))

# — GroupBy: Parent Education —————
print("\n===== BY PARENTAL EDUCATION =====")
print(df.groupby('parental level of
education').mean(numeric_only=True).round(2))
```

```

# — GroupBy: Test Preparation —————
print("\n===== BY TEST PREPARATION =====")
print(df.groupby('test preparation course').mean(numeric_only=True).round(2))

# — Correlation Matrix —————
print("\n===== CORRELATION MATRIX =====")
corr = df[['math score', 'reading score', 'writing score']].corr()
print(corr.round(4))

# — Heatmap —————
plt.figure(figsize=(6, 4))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.title("Correlation Heatmap – Exam Scores")
plt.tight_layout()
plt.savefig("outputs/heatmap.png", dpi=150)
plt.show()
print("Heatmap saved to outputs/heatmap.png")

```

Sample output:

```

Shape: (990, 8)
   gender race/ethnicity parental level of education      lunch \
0  female      group B      bachelor's degree      standard
1  female      group C              some college      standard
2  female      group B      master's degree      standard
3   male      group A      associate's degree  free/reduced
4   male      group C              some college      standard

   test preparation course  math score  reading score  writing score
0                none        72.0        72.0        74
1          completed        69.0        90.0        88
2                none        90.0        95.0        93
3                none        47.0        57.0        44
4                none        76.0        78.0        75

===== BY GENDER =====
      math score  reading score  writing score
gender
female      64.34        73.21        73.15
male        68.76        65.49        63.34

===== BY RACE/ETHNICITY =====
      math score  reading score  writing score
race/ethnicity
group A         61.63         64.67         62.67
group B         64.64         68.14         66.52
group C         64.80         69.34         68.12
group D         67.53         70.18         70.30
group E         73.82         73.03         71.41

```

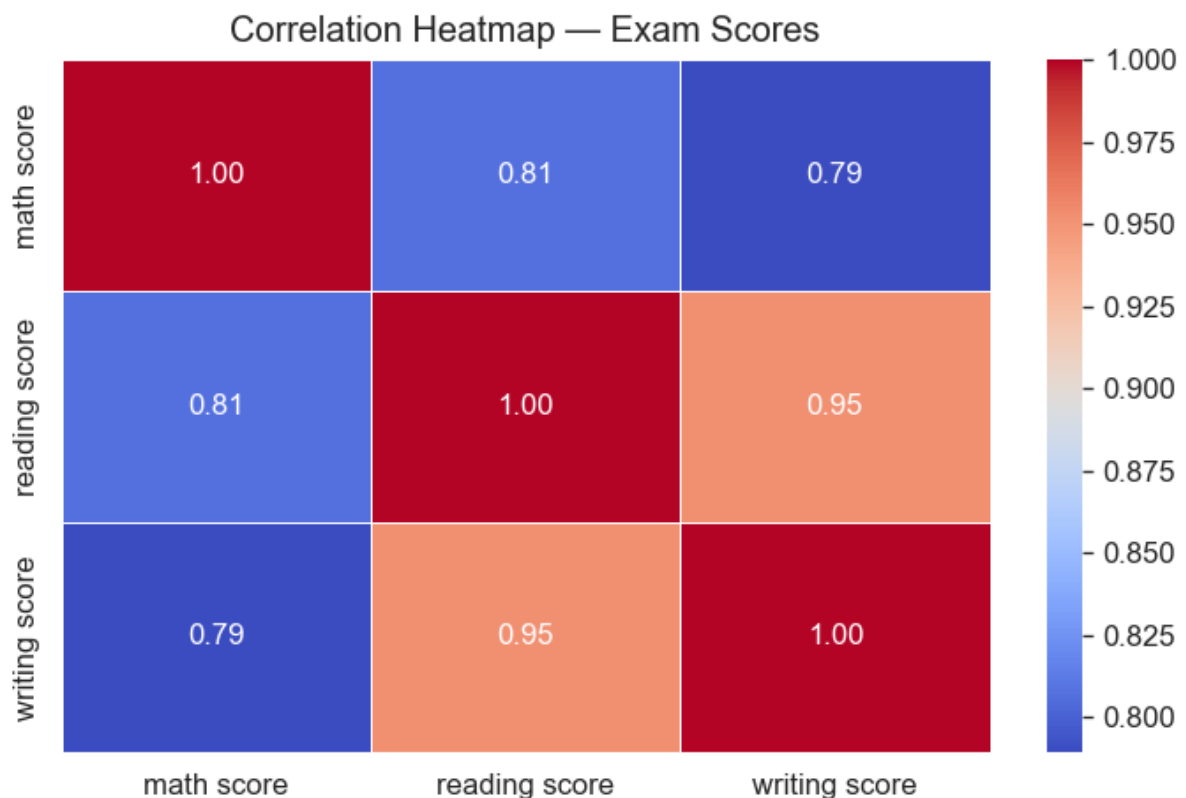
```

===== BY PARENTAL EDUCATION =====
               math score  reading score  writing score
parental level of education
associate's degree          68.06          71.05          70.00
bachelor's degree          69.39          73.00          73.38
high school                62.60          65.00          62.79
master's degree            69.75          75.37          75.68
some college               67.54          69.74          69.17
some high school           64.34          67.54          65.62

===== BY TEST PREPARATION =====
               math score  reading score  writing score
test preparation course
completed                  69.83          73.98          74.53
none                      64.59          66.88          64.90

===== CORRELATION MATRIX =====
               math score  reading score  writing score
math score          1.0000          0.8070          0.7893
reading score        0.8070          1.0000          0.9526
writing score        0.7893          0.9526          1.0000

```



Insights:

EDA Insights — Student Performance in Exams

1. Gender Differences: A clear pattern emerges across gender lines. Female students consistently outperform male students in reading and writing scores — by approximately 7 and 9 points respectively. However, male students hold a slight advantage in math scores, averaging about 5 points higher. This suggests that verbal and writing skills favor female students in this dataset, while quantitative performance is slightly stronger among males.

2. Race and Ethnicity: Students in group E consistently score the highest across all three subjects (math ~73.8, reading ~73.0, writing ~71.4), while students in group A score the lowest (math ~61.6, reading ~64.7, writing ~63.0). This could reflect underlying socioeconomic differences among groups and warrants further investigation with additional demographic variables.

3. Parental Education Impact: The relationship between parental education level and student performance is strongly positive. Students whose parents have a master's degree score the highest on average across all subjects, while students whose parents have only some high school education score the lowest. This highlights the important role that household educational background plays in student achievement — likely through factors like access to resources, study habits, and academic expectations at home.

4. Test Preparation Course: Students who completed the test preparation course show meaningful improvement across all subjects — approximately 5.6 points in math, 7.1 in reading, and 8.1 in writing compared to students who did not complete the course. This confirms that structured preparation significantly benefits academic outcomes, especially in writing tasks.

5. Correlation Between Subjects: Reading and writing scores show a very strong positive correlation ($r \approx 0.95$), which makes intuitive sense — both are language-based skills that reinforce each other. Math has a moderate correlation with both reading ($r \approx 0.82$) and writing ($r \approx 0.80$), suggesting that strong general academic ability links all three subjects, though language and math skills are still somewhat distinct skill sets.

Summary: Overall, parental education level and test preparation course completion are the two strongest predictors of student performance in this dataset. Gender differences are present but more subject-specific. These insights can guide educational policy decisions around parental engagement programs and the expansion of test preparation resources.

Task 5 — Data Visualization

Source Code:

```
import os
os.makedirs('outputs', exist_ok=True)

# — Load Cleaned Data —————
df = pd.read_csv("cleaned_students.csv")
avg_math = df.groupby('parental level of education')['math score'].mean()

# — Plot 1: Histogram with KDE —————
plt.figure(figsize=(6, 4))
sns.histplot(df['math score'], kde=True, color='steelblue')
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")
plt.grid(True)
plt.tight_layout()
plt.savefig("outputs/plot1_histogram.png", dpi=150)
plt.show()
print("Plot 1 saved.")

# — Plot 2: Boxplot —————
```

```

plt.figure(figsize=(6, 4))
sns.boxplot(x='gender', y='math score', data=df,
            palette={'male': 'steelblue', 'female': 'salmon'})
plt.title("Math Score by Gender")
plt.xlabel("Gender")
plt.ylabel("Math Score")
plt.grid(True)
plt.tight_layout()
plt.savefig("outputs/plot2_boxplot.png", dpi=150)
plt.show()
print("Plot 2 saved.")

# — Plot 3: Scatter Plot —————
plt.figure(figsize=(6, 4))
sns.scatterplot(x='reading score', y='writing score',
                hue='gender', data=df,
                palette={'male': 'steelblue', 'female': 'salmon'},
                alpha=0.6)
plt.title("Reading vs Writing Scores")
plt.xlabel("Reading Score")
plt.ylabel("Writing Score")
plt.grid(True)
plt.tight_layout()
plt.savefig("outputs/plot3_scatter.png", dpi=150)
plt.show()
print("Plot 3 saved.")

# — Plot 4: Bar Chart —————
plt.figure(figsize=(8, 4))
avg_math.sort_values().plot(kind='bar', color='teal', edgecolor='white')
plt.title("Average Math Score by Parent Education")
plt.xlabel("Parent Education")
plt.ylabel("Average Math Score")
plt.xticks(rotation=30, ha='right')
plt.grid(True, axis='y')
plt.tight_layout()
plt.savefig("outputs/plot4_bar.png", dpi=150)
plt.show()
print("Plot 4 saved.")

# — Combined 2x2 Figure —————
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle("Student Performance Analysis", fontsize=15, fontweight='bold')

# Histogram
sns.histplot(df['math score'], kde=True, color='steelblue', ax=axes[0, 0])
axes[0, 0].set_title("Math Score Distribution")
axes[0, 0].set_xlabel("Math Score")

```

```

axes[0, 0].set_ylabel("Frequency")
axes[0, 0].grid(True)

# Boxplot
sns.boxplot(x='gender', y='math score', data=df,
            palette={'male': 'steelblue', 'female': 'salmon'}, ax=axes[0, 1])
axes[0, 1].set_title("Math Score by Gender")
axes[0, 1].set_xlabel("Gender")
axes[0, 1].set_ylabel("Math Score")
axes[0, 1].grid(True)

# Scatter
sns.scatterplot(x='reading score', y='writing score', hue='gender',
               data=df, palette={'male': 'steelblue', 'female': 'salmon'},
               alpha=0.6, ax=axes[1, 0])
axes[1, 0].set_title("Reading vs Writing Scores")
axes[1, 0].set_xlabel("Reading Score")
axes[1, 0].set_ylabel("Writing Score")
axes[1, 0].grid(True)

# Bar chart
avg_math.sort_values().plot(kind='bar', color='teal',
                            edgecolor='white', ax=axes[1, 1])
axes[1, 1].set_title("Avg Math by Parent Education")
axes[1, 1].set_xlabel("Parent Education")
axes[1, 1].set_ylabel("Average Math Score")
axes[1, 1].tick_params(axis='x', rotation=30)
axes[1, 1].grid(True, axis='y')

plt.tight_layout()
plt.savefig("best_visualization.png", dpi=150)
plt.show()
print("Combined figure saved to best_visualization.png")

```

Sample Output:

